

General Solutions

①

In ODEs, an IVP is often solved by first finding a general solution that depends on an arbitrary constant and then using the initial condition to evaluate that constant.

For quasilinear 1st order PDEs, a similar process may be achieved by the method of Lagrange, which produces a general solution that depends on an arbitrary function.

We demonstrate the method with an example:

(2)

Example 4 (Revised): Find a general solution of $u u_x + y u_y = x$.

Step 1: Find a function $\phi(x, y, z)$ such that the level surface

$$\phi(x, y, z) = \text{const}$$

is an integral surface of the vector field $V = \langle a, b, c \rangle$.

i.e. ϕ is constant along the characteristics:

$$\frac{dx}{dt} = a, \quad \frac{dy}{dt} = b, \quad \frac{dz}{dt} = c,$$

and so differentiating

$$\phi(x(t), y(t), z(t)) = \text{const}$$

w.r.t. 't' shows that ϕ satisfies

$$a \phi_x + b \phi_y + c \phi_z = 0.$$

We write the characteristics in the ^③ nonparametric form:

$$\frac{dx}{a} = \frac{dy}{b} = \frac{dz}{c}$$

In this example:

$$\frac{dx}{z} = \frac{dy}{y} = \frac{dz}{x}$$

Consider the first and third expressions that does not contain a 'y' term:

$$\frac{dx}{z} = \frac{dz}{x}$$

$$\Rightarrow x dx - z dz = 0$$

$$\Rightarrow \frac{x^2}{2} - \frac{z^2}{2} = c$$

or

$$z^2 = x^2 + c_1$$

$$\phi(x, y, z) \equiv z^2 - x^2 = c_1$$

④
The function $\phi = z^2 - x^2$ is clearly constant along the characteristic curve; in fact it is easy to verify that

$$z \phi_x + y \phi_y + x \phi_z = 0.$$

Thus we have found one function:

$$\phi = z^2 - x^2.$$

Step 2: Find a second independent solution $\psi(x, y, z) = \text{const.}$

For that let us use $\phi = z^2 - x^2 = C_1$ to write $z = \sqrt{x^2 + C_1}$.

Consider
$$\frac{dx}{\sqrt{x^2 + C_1}} = \frac{dy}{y}.$$

$$\Rightarrow \ln |x + \sqrt{x^2 + C_1}| + C_2 = \ln |y|$$

(5)

$$\Rightarrow \left| \frac{y}{x + \sqrt{x^2 + C_1}} \right| = C_2$$

Therefore $\psi(x, y, z) \equiv y(x+z)^{-1}$

is constant. It is easy to

check that $z\psi_x + y\psi_y + x\psi_z = 0$

Thus we have found a second independent solution.

Step 3: The general solution can then be written as

$$F\left(u^2 - x^2, \frac{y}{x+u}\right) = 0,$$

where $F(\phi, \psi)$ is an arbitrary C^1 function.

Or we can write $\phi = f(\psi)$, with f an arbitrary C^1 function.

In this example, $u^2 = f\left(\frac{y}{x+u}\right) + x^2$ is the general solution.

Example 5: (Inviscid Burgers Eqn) (6)

Derivation of the PDE: $u u_x + u_t = 0$.

This PDE models the flow of a stream of particles along the x -axis. (1 dimensional model).

Let us assume that each particle has constant velocity. We may interpret this as a velocity field

$u(x, t)$: velocity at a position x and at time t .

If we follow an individual particle, we get a function $x(t)$ for which $u(x(t), t)$ remains constant.

$$\frac{d}{dt} u(x(t), t) = 0$$

$$u u_x + u_t = 0 \quad (\text{because here the velocity } \frac{dx}{dt} = u).$$

(7)

Suppose the velocity field is known initially : $u(x,0) = h(x)$.

Then we can solve the PDE with this initial data profile using the method of characteristics:

Step 1: $\Gamma \rightarrow (s, 0, h(s))$.

Step 2: characteristic equations:

$$\frac{dx}{dt} = z, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = 0$$

Step 3: $\frac{dy}{dt} = 1 \Rightarrow y = t + C_1(s)$
 $0 = y(s,0) = C_1(s)$

$$\therefore \boxed{y = t}$$

$$\frac{dz}{dt} = 0 \Rightarrow z = C_2(s)$$

$$h(s) = z(s,0) = C_2(s).$$

$$\therefore \boxed{z = h(s)}$$

$$\frac{dx}{dt} = z \Rightarrow \frac{dx}{dt} = h(s) \Rightarrow x(s,t) = h(s)t + C_3(s)$$

$$s = x(s,0) = C_3(s).$$

(8)

$$\therefore \boxed{x = h(s)t + s}$$

$$\Downarrow$$

$$s = x - zy.$$

Step 4: Finally $z = h(s) = h(x - zy)$
or

$u(x, y) = h(x - uy)$ is the solution given implicitly.

Alternative method is by method of Lagrange:

Step 1: Find a function $\phi(x, y, z)$.

For that consider the characteristics:

$$\frac{dx}{z} = \frac{dy}{1} = \frac{dz}{0}$$

The last expression implies that z is constant along the characteristics, so we can take $\phi(x, y, z) = z$.
(clearly $u\phi_x + \phi_y = 0$).

(9)

Step 2: Find a second independent solution $\psi(x, y, z) = \text{const.}$

Consider the first two expressions:

$$\frac{dx}{z} = \frac{dy}{1}.$$

Treating z as a constant from step 1,

$$x = zy + C_2.$$

Now take $\psi(x, y, z) = x - zy$, we find that ψ is constant along characteristics.

Step 3: The general solution is written as:

$z = f(x - zy)$, which defines u implicitly.
The initial condition shows that $f(\psi) = h(\psi)$.

— x —